Mukul Nayak

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.50
RAMAKRISHNA VIVEKANANDA SARADA VIDYAPITH	Class XII (ISC)	2019	88.60%
ST. JOSEPH CONVENT HIGH SCHOOL	Class X (ICSE)	2017	88.40%

PROFESSIONAL EXPERIENCE

SPIKEWELL | SDE Intern | *Jan 2024 - Jun 2024*

- **Improved classification performance by 10%** over the existing model on a **highly imbalanced dataset**.
- **Built an ETL pipeline** for automated API-to-**Azure** data ingestion, enabling iterative model training and **CI/CD**; deployed the trained model as a **web service**.
- **Worked with Python, Azure and Power BI** across machine learning, cloud deployment and data analytics workflows.

PROJECTS

PARALLEL CONNECTED COMPONENTS | 2026 | **Guided by Prof. Sathya Peri**

- Engineered **parallel Hybrid BFS + Shiloach-Vishkin and lock-free DSU algorithms** using **multithreading, CAS, path compression and contention-aware synchronization**.
- Benchmarked **1-12 thread scalability** on graphs up to **10K nodes**, varying density and component structure; analyzed **runtime, speedup and memory/cache contention**.

LOW-LATENCY OPTIONS TRADING & EXECUTION ENGINE | 2026

- **Designed and implemented** an event-driven C++ trading engine with market-data processing, order-book management, spread strategy, risk checks, and simulated execution.
- **Implemented lock-free SPSC queues** with preallocated storage, acquire-release atomics, and cache-line alignment for low-latency market-data processing.
- **Developed latency and performance benchmarks** measuring throughput and **P50/P95/P99/P99.9** latency across the trading pipeline.

CLOUD-NATIVE DEVOPS & GITOPS | 2022 | **Guided by Mr. Sumit Shah**

- Built an **end-to-end CI/CD pipeline** with **Jenkins, Docker, Amazon ECR and ArgoCD**, automating application build and GitOps-based deployment.
- Provisioned **7+ AWS infrastructure components** using **Terraform**, including **VPC, EKS, ECR, EFS, EC2, Route 53 and ACM**.
- Implemented **Prometheus, Grafana and AlertManager** for centralized monitoring and automated alerting across the **Kubernetes deployment**.

SKILLS

Programming: C, C++, Cuda, SQL, Linux

Parallel & High-Performance Computing: OpenMP, Parallel Programming, GPU Programming, Performance Optimization

Machine Learning: NumPy, Pandas, PyTorch, Python

Cloud & Development: Git, Linux, Azure, AWS, REST APIs

RELEVANT COURSES

Mathematics : Probability and Stochastic Processes, Linear Algebra, Topics in Optimization

AI/ML : Foundations of Machine Learning, Deep Learning

Computer Science : Advanced DSA, Parallel and Concurrent Programming

AWARDS AND ACHIEVEMENTS

2023 Global **Rank 150** Codechef Starters 84

2018 State **Rank 8** International Informatics Olympiad

2017 **Rank 1** Gold medalist, Prabhat Khabar - Rank 1 among ICSE schools in the town

INTERESTS

Competitive Programming, Chess and Cricket